a Primary reason recorded for efficacy failure in the 27 classes whose target engagement was confirmed in patients

Target hypothesis: the target and the direction of the intended effect n = 16

◆ Not causal: output moved, disease did not (12)
- H3 antagonists · AD
- 5-HT6 antagonists · AD
- anti-IL-13 · asthma
- anti-IL-33 · atopic eczema
- p38 inhibitors · COPD
- HCAR2 agonists · lipids
- HDAC inhibitors · MDS
- anti-IGF-1R · NSCLC
- PDE10A inhibitors · SCZ
- GPR119 agonists · T2D
- 11β-HSD1 inh. · T2D
- MEK inhibitors · pancreas

◆ Causal at another disease stage (2)
- BACE1 inhibitors · AD
- CDK4/6 inhibitors · breast

◆ Causal only in a subgroup (0)
none

Causal but compensated or bypassed (1)
- anti-EGFR (new) · CRC

Causal, effect too small (1)
- pan-PI3K inhibitors · breast

Modulation: the drug, its dose, or the site of engagement n = 6

Insufficient degree or duration of engagement (6)
- HSP90 inhibitors · breast
- PI3K/mTOR inh. · breast
- NRTIs (new) · HIV
- anti-GM-CSF · RA
- BTK inhibitors · RA
- proteasome inh. · myeloma

Trial: endpoint, population, or treatment context n = 4

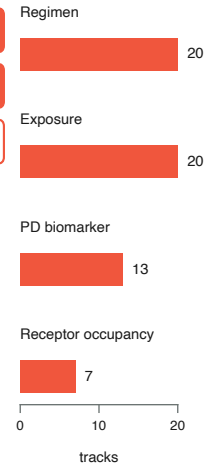
Population or placebo response (3)
- allergen peptides · allergy
- allergen IT · rhinitis
- anti-CGRP · migraine

Combination or comparator context (1)
- DP2 antagonists · asthma

Not stated in the record n = 1

Not stated in the record (1)
- DP2 antagonists · rhinitis

b Dose tracks scored "reached, not enough" by target reached (n = 60)



class never approved in this disease

class approved in this disease; the contradiction is a stage, a population, or one member

◆ category present in the current manuscript draft; the remaining categories are added here